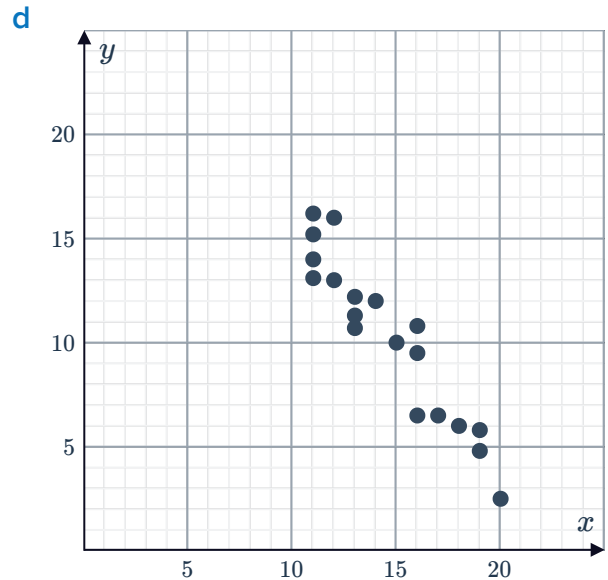
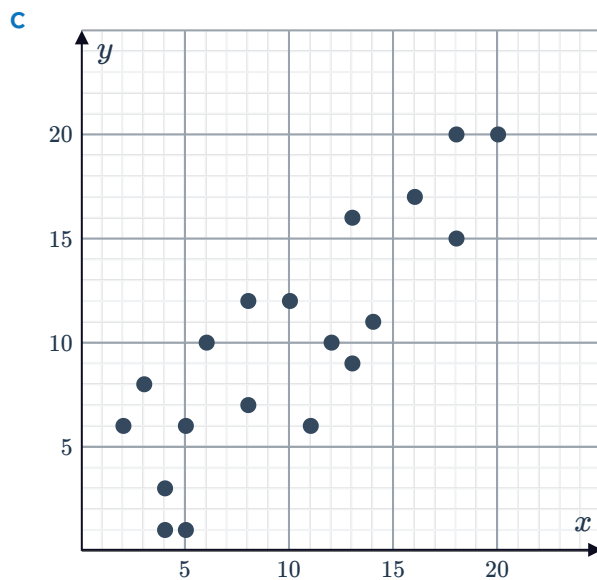
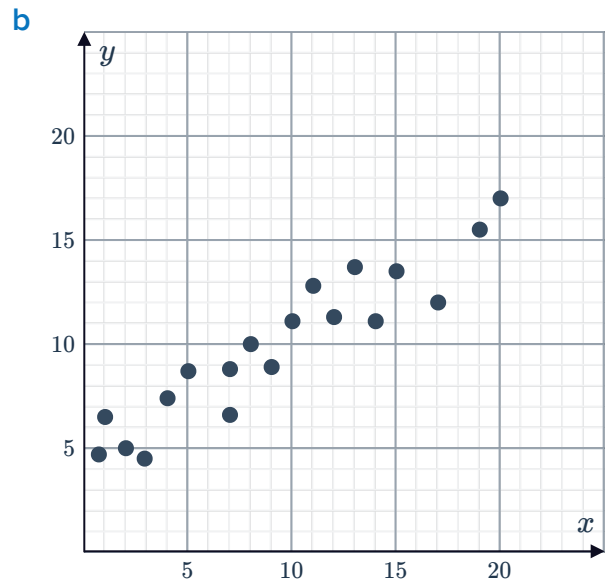
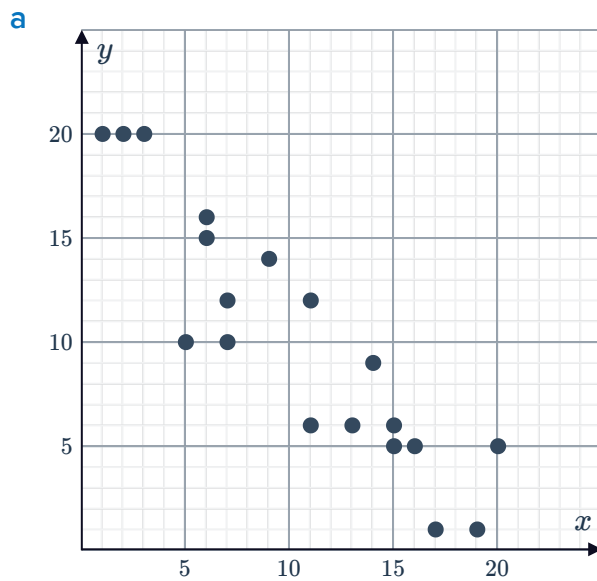


Line of best fit

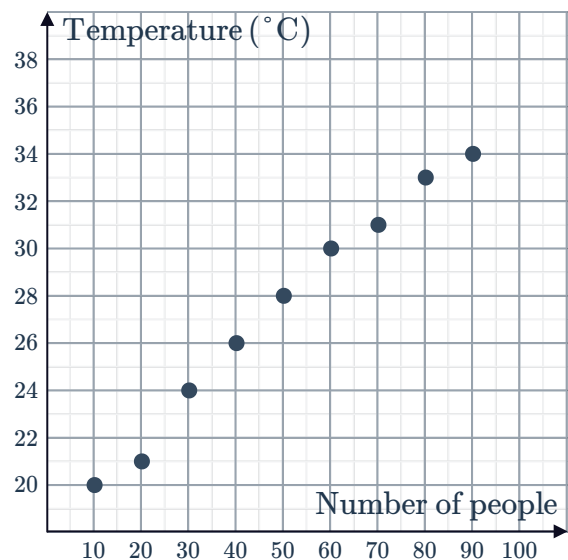


1 Draw an approximate line of best fit by hand for each of the the scatter plots below:

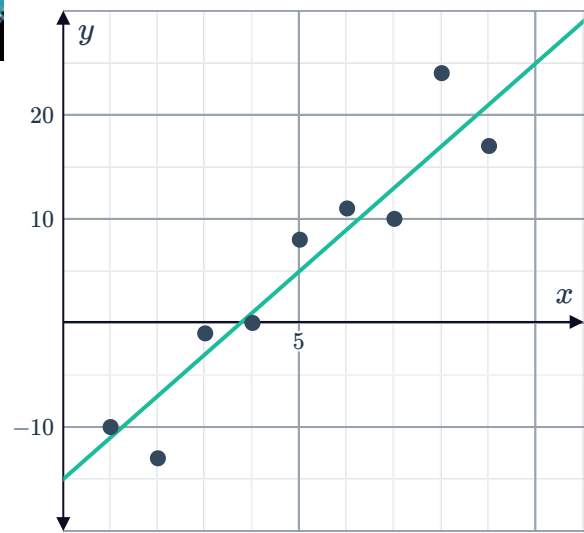


2 The following scatter plot displays data for the number of people in a room and the room temperature, as collected by a researcher:

- Find the mean point.
- Draw a line of best fit through this mean point.



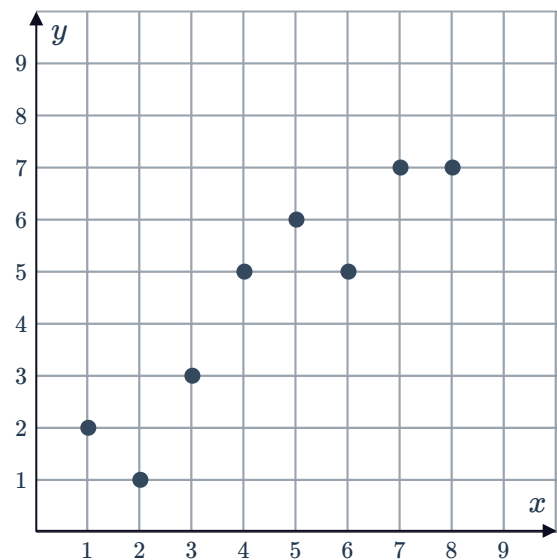
- 3 Find the equation of the line of best fit for the data shown:



- 4 The equation for the line of best fit is given by $P = -4t + 116$, where t is time. Determine if the relationship shows that over time, P is increasing, decreasing or remain constant.

- 5 The following scatter plot shows the data for two variables, x and y :

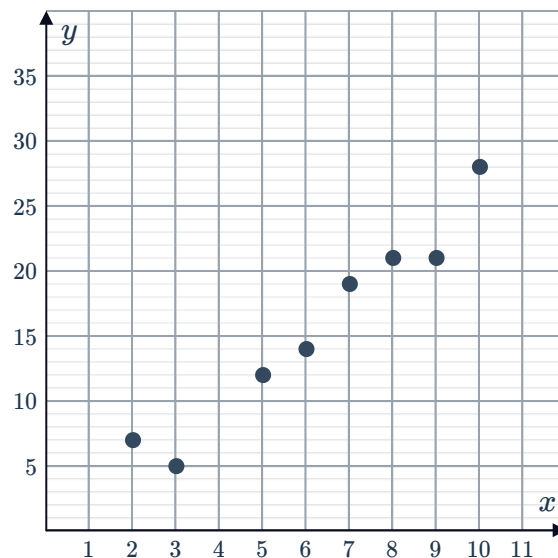
- a Sketch the line of best fit for this data.
- b Use your line of best fit to estimate the value of y when:
- i $x = 4.5$ ii $x = 9$



6 Consider the data plotted on the following scatter plot:

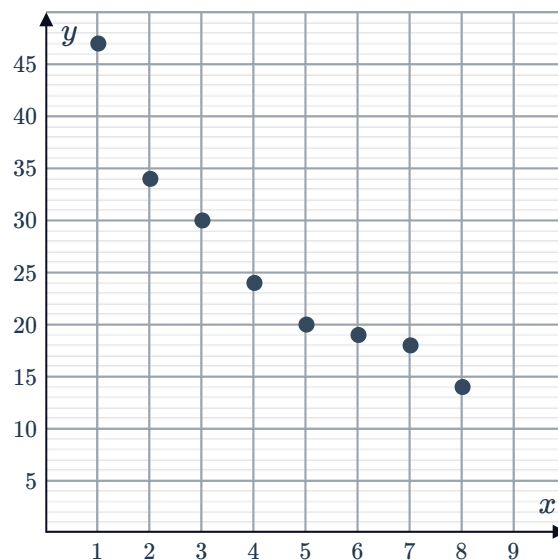


- a Is there a negative or positive relationship between the x and y variables?
- b Sketch the line of best fit for this data.
- c Find the equation of your line of best fit.



7 Consider the data plotted on the following scatter plot:

- a Is there a negative or positive relationship between the x and y variables?
- b Sketch the line of best fit for this data.
- c Find the equation of your line of best fit.



8 A least squares regression line is given by $y = 3.59x + 6.72$.

- a State the gradient of the line.
- b Does the gradient of the line indicate that the bivariate data set has a positive or negative correlation?
- c As x increases by 1 unit, how does the value of y change?
- d State the value of the y -intercept.

9 A least squares regression line is given by  $y = -3.67x + 8.42$.

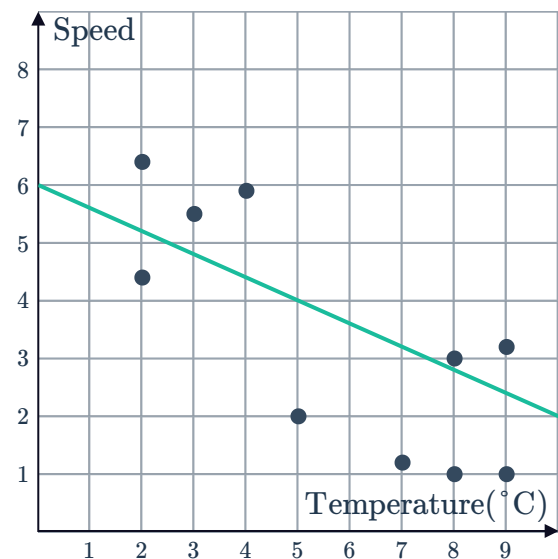
- a State the gradient of the line.
- b Does the gradient of the line indicate that the bivariate data set has a positive or negative correlation?
- c As x increases by 1 unit, how does the value of y change?
- d State the value of the y -intercept.

10 A least squares regression line is given by $T = -7.26S + 4.35$.

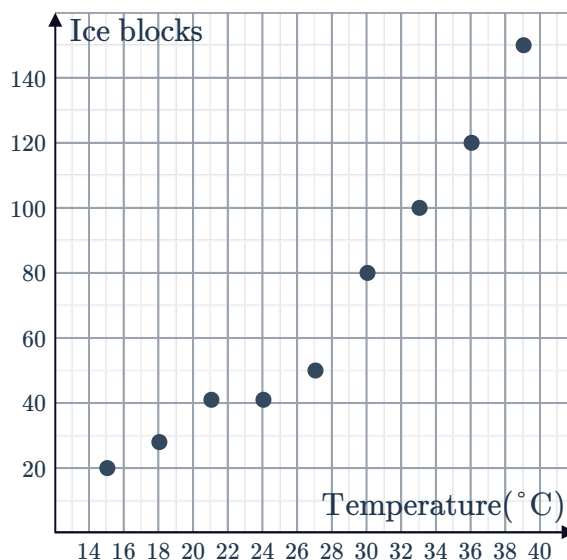
- a State the gradient of the line.
- b Does the gradient of the line indicate that the bivariate data set has a positive or negative correlation?
- c As x increases by 1 unit, how does the value of y change?
- d State the value of the vertical intercept.

Applications

- 11 The average monthly temperature and the average wind speed, in knots, in a particular location was plotted over several months. The graph shows the points for each month's data and their line of best fit: Use the line of best fit to approximate the wind speed on a day when the temperature is 5°C .



- 12 The following scatter plot graphs data for the number of ice blocks sold at a shop on days with different temperatures.



- Sketch the line of best fit for this data.
- Use your answer line of best fit to estimate the number of ice blocks that will be sold on a:
 - 31°C day
 - 42°C day
- Does the number of ice blocks sold increase or decrease as the temperature increases?

- 13 Jordano measures his heart rate at various times while running. The data is shown below in the table:

Time (minutes)	0	2	4	6	8	10
Heart Rate (BPM)	58	54	58	67	67	74

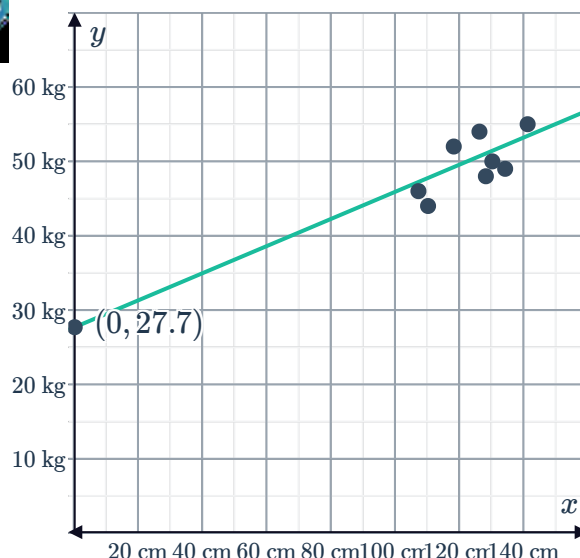
- Plot the data and sketch the line of best fit.
- Which data point is closest to the line of best fit?
- Which data point is furthest from the line of best fit?

- 14 A dam used to supply water to the neighboring town had the following data recorded for its volume over a number of months:

Month	1	2	3	4
Volume (billion of litres)	116	106	104	92

- Plot the data on a number plane.
- Sketch the line of best fit for this data.

- 15 The heights (in cm) and the weights (in kg) of 8 primary school children is shown on the scattergraph below:



- 16 The amount of money households spend on dining out each week, D , is measured against their weekly income, I .

The following linear regression model is fitted to the data: $D = 0.3I + 27$.

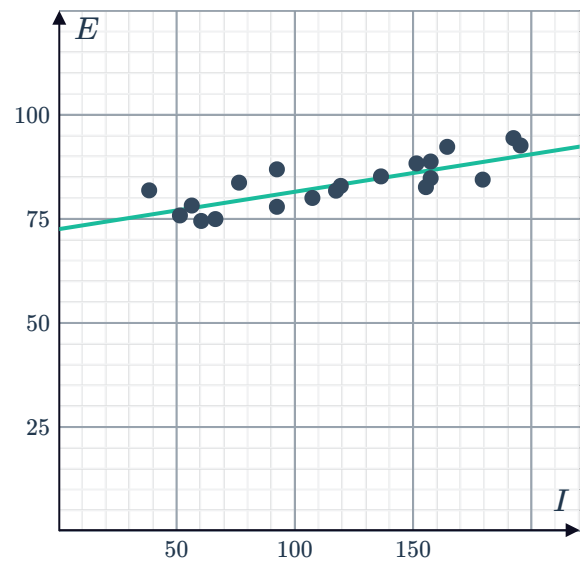
- Explain the meaning of the vertical intercept.
- Does the interpretation in the previous part make sense in this context? Explain your answer.
- State the gradient of the line.
- If the weekly income of a family increases by \$200, by how much can we expect their spending on dining out to increase?

- 17 The number of hours spent watching TV each evening, h , is measured against the percentage results, m , achieved in the Economics exam.

The following linear regression model is fitted to the data: $m = -10h + 97$.

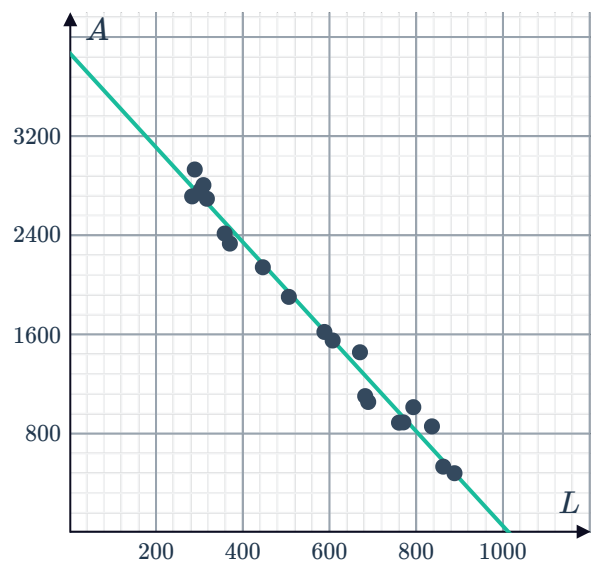
- Explain the meaning of the vertical intercept.
- Does the interpretation in the previous part make sense in this context? Explain your answer.
- State the gradient of the line.
- If a student increases the amount of TV they watch by 3.5 hours, by how much can we expect their Economic exam mark to drop?

- 18** The life expectancy (E), in years, of individuals at different annual incomes (I), per \$1000, is shown:
The equation of the line of best fit is $E = 0.09I + 72.55$.



- By how much does average life expectancy change for each \$1000 of annual income?
- Find the average life expectancy of someone who earns no income.

- 19** Scientists record the number of aphids (A) in areas with different numbers of ladybeetles (L), and graph the data in the scatterplot shown:
The equation of the line of best fit is $A = -3.82L + 3865.21$.



- How much does the average aphid population change by with each extra ladybeetle? Give your answer to the nearest aphid.
- Find the average aphid population of a region with no ladybeetles. Give your answer to the nearest aphid.